Fiscal Year:	FY 2017	**Task Last Updated:**	FY 01/16/2017
PI Name:	Wolverton, Scot Ph.D.		
Project Title:	Characterizing Plant Gravity Perception Systems		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	(1) Developmental Biology		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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Zip Code:	43015-2333	**Congressional District:**	12
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2014 Space Biology Flight (ILSRA) NNH14ZTT002N
Start Date:	03/16/2015	**End Date:**	03/15/2019
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	5
No. of Bachelor's Candidates:	5	**Monitoring Center:**	NASA KSC
Contact Monitor:	Levine, Howard	**Contact Phone:**	321-861-3502
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Flight Program:	ISS		
Flight Assignment:	NOTE: Extended to 3/15/2019 per NSSC information (Ed., 12/18/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX15AG55G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Plant growth is characterized by plasticity, allowing environmental cues like light and gravity to play a significant role in determining the overall form of the plant body. Statolith sedimentation in specialized cells represents one way that plants perceive gravity. Plants lacking functional statoliths show reduced but consistent gravity responses, suggesting the presence of a non-statolith sensory system. This proposal seeks to investigate the gravity perception machinery in roots of several genotypes of the higher plant Arabidopsis thaliana by taking advantage of the unique microgravity environment of Earth orbit on the International Space Station (ISS). One thrust of the work described here is to characterize the physiological properties of the putative non-statolith sensory system, including the threshold acceleration force required to activate it, and its capacity to modulate downstream growth regulation machinery. The second thrust is to determine whether the two gravity sensory systems control a single or multiple output systems. Addressing this question requires the microgravity environment in order to quantify the kinetics of gravitropism in roots having or lacking functional statoliths under fractional g conditions. The overall outcome of the work proposed here will be an improved understanding of how higher plants perceive gravity and transduce this physical cue into directional growth, a fundamental requirement for improving plants for production on Earth and in long-term spaceflight missions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The overall outcome of the work proposed here will be an improved understanding of how higher plants perceive gravity and transduce this physical cue into directional growth, a fundamental requirement for improving plants for production on Earth and in long-term spaceflight missions.
Task Progress:	Plants are extremely sensitive to gravity and other environmental cues, which they use to direct their growth. Even though scientists have been studying how plants sense and respond to gravity for over 100 years, we still do not fully understand these processes. One way plants detect gravity is through the falling of dense bodies inside of cells called statoliths. But plants that lack statoliths because of a genetic mutation can still partially detect gravity, suggesting there's at least one other way that plants can sense gravity. We plan to study these mutant plants in space, where there is almost no gravity, and add back small amounts of gravity by growing them on a centrifuge. This will allow us to better understand how the mutants are detecting and responding to gravity compared with normal plants. This year, which was our second year of funding for this project, we continued to define our experimental plan for our spaceflight experiment. We resolved the last of a number of questions that needed to be answered before our project can proceed toward flight implementation. We also began building a data set of gravity responses with plants growing under the same conditions as the flight experiments. We finished the last of our preliminary experiments, in which we determined the germination rates of seeds stored in flight conditions for many months, so we can determine how long we can safely store our experiment before launch and while onboard the Space Station before our seeds begin to get too old to germinate and grow normally. We found that seeds attached to flight membranes and stored for even 8 months remain highly viable, with germination rates at or above 90%, as long as the seeds were not glued with an excessive amount of guar gum. The excess gum dramatically reduced germination rates to below 50%, revealing an important parameter in properly setting up the experiment prior to launch. We carried out a Science Verification Test this year too, the purpose of which is to demonstrate that the plants could grow and respond as required in the same hardware used for the flight experiment. We performed this experiment as much like the flight experiment as possible, and we found that the germination, growth, and response to directional light treatments all fell within the "Excellent" range as established previously for each of these parameters. We also began planning for an Experiment Verification Test, which will take place in a growth chamber that mimics the atmosphere and lighting conditions available on the Space Station. In addition to these experiments to prepare for flight, we also started studying the gravity response pathway in a number of other mutants, to test whether some of these mutations might give insight into the non-statolith gravity sensing system. To perform these experiments, we use a special device that we made just for measuring gravity responses in plants and that lets us hold the plant very still in a specific position relative to gravity. We are carrying out these experiments in the lab on a continuous basis, looking at many different mutants for clues to gravity sensing in plants. Our homemade device is approaching the end of its useful life, so this year we began work to update and modernize this instrument using off-the-shelf electronic components rather than the previous, custom-made components.
Bibliography Type:	Description: (Last Updated: 05/04/2026)
Abstracts for Journals and Proceedings	Sarkel E, Wolverton C, Muday G. "Reactive Oxygen Species and Flavonols Modulate the Root Gravitropic Response." Annual Meeting of the American Society of Plant Biologists, Plant Biology 2016, Austin, Texas, July 9-13, 2016. Plant Biology 2016, Austin, Texas, July 9-13, 2016. , Jul-2016
Abstracts for Journals and Proceedings	George M, Quick R, Wojnoski A, Madonich N, Wolverton C. "Gravitropism and Long-term Seed Storage on PES Membranes in Preparation for a Flight Experiment." 32nd Annual Meeting of the American Society for Gravitational and Space Research, Cleveland, OH, October 26-29, 2016. 32nd Annual Meeting of the American Society for Gravitational and Space Research, Cleveland, OH, October 26-29, 2016. , Oct-2016